

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

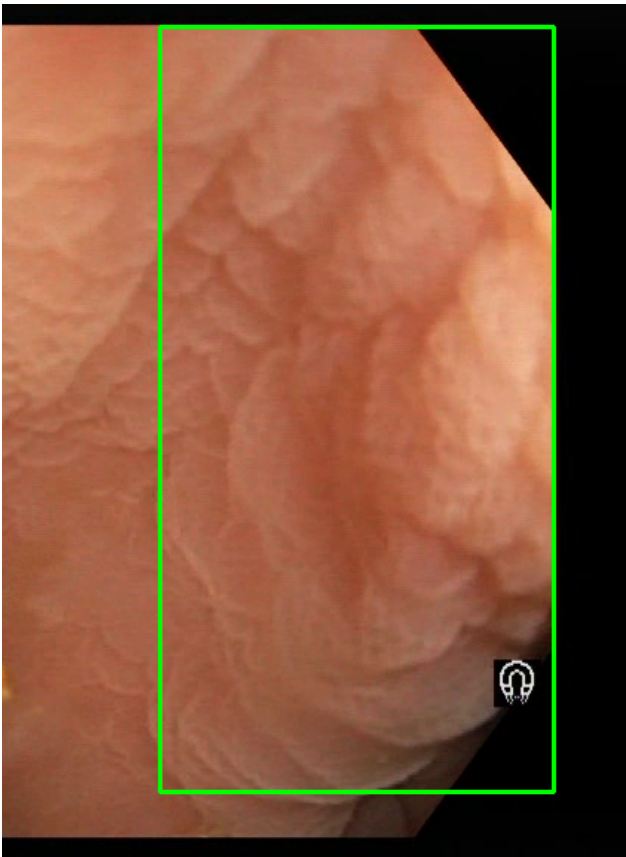
Video ID:	31573497-b434-4c27-86ba-a53a65d5f9ce
Total Frames:	15006
Frame Rate:	25.0 fps (assumed)
Duration:	600.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	246	✓
RT-DETR Detections	595	✓
MedSAM2 Detections	711	✓
Consensus (3-Model Agreement)	4	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	4	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 77 frames)



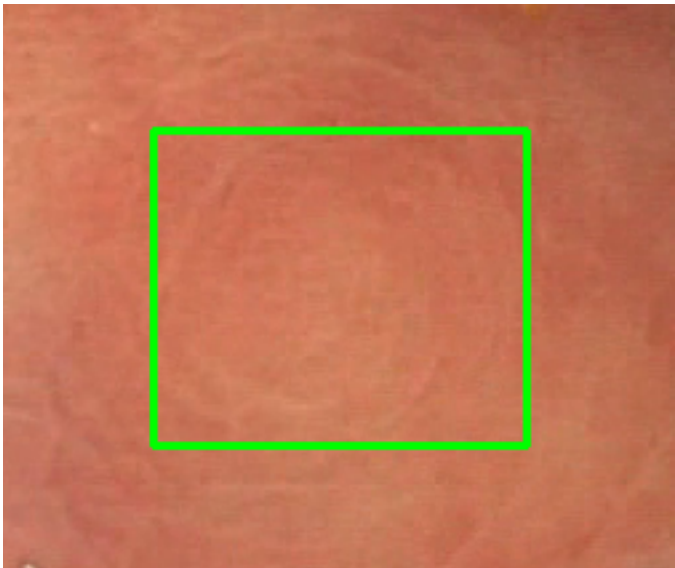
Feature	Value	Interpretation
Frame Index	8425	Frame 8425 of 15006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.208	Color-based vascularization
Relative Radius	37.6%	Polyp size as % of frame diagonal
Texture Score	0.189	Surface morphology
Vessel Visibility	0.721	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	MALIGNANT_PO LYP	High suspicion for malignancy — biopsy and urgent resection recommended
Detection Confidence	73.7%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #2 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 23 frames)



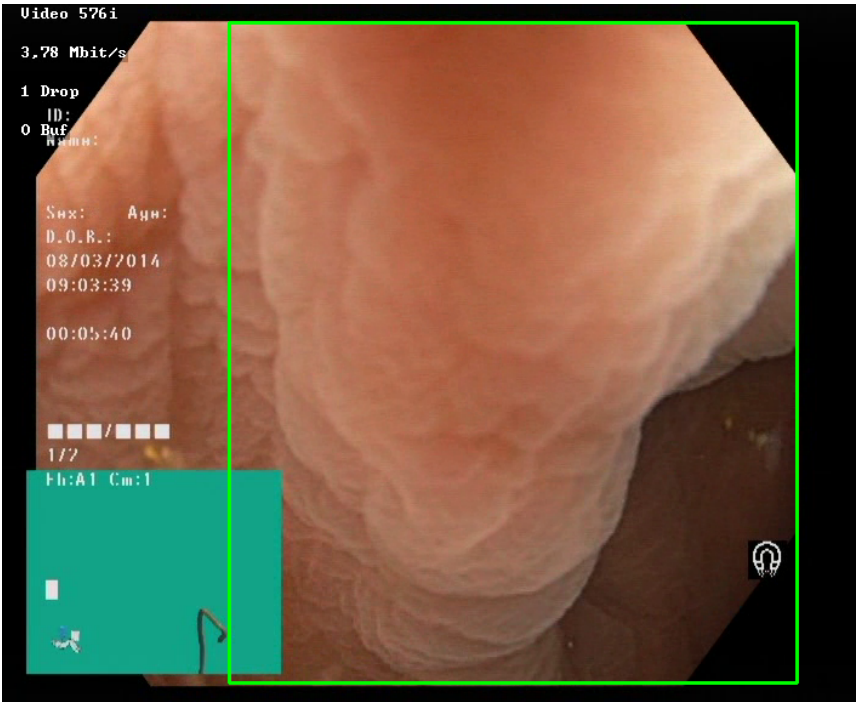
Feature	Value	Interpretation
Frame Index	5832	Frame 5832 of 15006
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_PO LYP	Type confidence: 92.0%
Redness Score	0.213	Color-based vascularization
Relative Radius	18.9%	Polyp size as % of frame diagonal
Texture Score	0.017	Surface morphology
Vessel Visibility	0.988	Vascularization indicator

Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	72.6%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #3 — MALIGNANT_POLYP [HIGH_RISK] (Tracked across 20 frames)



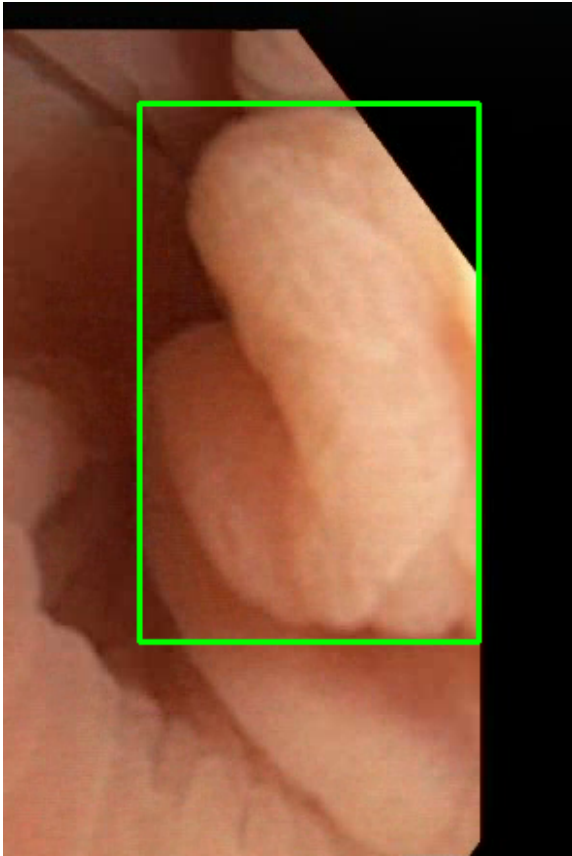
Feature	Value	Interpretation
Frame Index	8494	Frame 8494 of 15006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	MALIGNANT_POLYP	Type confidence: 92.0%

Redness Score	0.163	Color-based vascularization
Relative Radius	63.8%	Polyp size as % of frame diagonal
Texture Score	0.120	Surface morphology
Vessel Visibility	0.495	Vascularization indicator
Risk Tier	HIGH RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	72.8%	YOLO / RT-DETR / track confidence

Type Assessment: Morphology consistent with high-grade neoplasia — biopsy and urgent resection strongly recommended.

Low-confidence HIGH RISK detection - Further evaluation recommended

POLYP #4 — BLEEDING_POLYP [HIGH_RISK] (Tracked across 16 frames)



Feature	Value	Interpretation
Frame Index	6976	Frame 6976 of 15006
Detection Method	Detector Consensus (MedSAM2 Unavailable)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	BLEEDING_POLYP	Type confidence: 92.0%
Redness Score	0.194	Color-based vascularization
Relative Radius	6.6%	Polyp size as % of frame diagonal
Texture Score	0.023	Surface morphology
Vessel Visibility	0.974	Vascularization indicator
Risk Tier	HIGH_RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output

Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	72.4%	YOLO / RT-DETR / track confidence

Type Assessment: Active hemorrhagic lesion — elevated vascularity with redness signal. Immediate clinical review required.

Low-confidence HIGH RISK detection - Further evaluation recommended

CLINICAL IMPRESSION

Summary:

A total of 4 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.